

	to the earthed plate/clockwise	A1	[2]
7	(a) potential difference/current	B1	[1]
	(b) (i) 1) 1.13 W 2) 1.50 V	B1	[1]
	(ii) power = V^2 / R <u>or</u> power = VI and $V = IR$ $R = 1.50^2 / 1.13$ $= 1.99 \Omega$	C1 A1	[2]

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(iii) either $E = IR + Ir$ or voltage divided between R and r C1
 $I = 1.5 / 2.0 (=0.75 \text{ A})$ p.d. across $R =$ p.d. Across $r = 1.5$ C1
 $3.0 = 1.5 + 0.75r$
 $r = 2.0 \Omega$ so $R = r = 1.99 \Omega$ A1

[3]

(c) larger p.d. across R means smaller p.d. across r M1
 smaller power dissipation at larger value of V A1
 since power is VI and I is same for R and r A1

[3]